Jinming Xing

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Education

North Carolina State University **Sep 2023 – Present**

- PhD in Computer Science
- **Research Interest:** LLM Agents & Agentic Systems, LLM-Guided Graph Reasoning

Shenzhen University **Sep 2019 – Jun 2023**

- BS in Computer Science (Honored)
- **Coursework:** Probabilities, Linear Algebra, Data Structures and Algorithms, Computer Networks, Internet of Things, Cloud Computing, Database, Machine Learning, Practical Deep Learning, Computer Vision

Skills

- **LLM Agents & Agentic Systems:** MCP (server/client implementation, tool-schema design, prompts), tool calling, multi-agent orchestration (OpenAI/Claude Agents SDK, LlamaIndex Workflows), ReAct / plan-and-execute / reflection & self-critique loops, supervisor-subagent delegation, context engineering & long-horizon memory, human-in-the-loop checkpointing, agentic RAG, verifier-gated and self-critiquing generation
- **LLM Post-Training & Inference:** GRPO, PPO, DPO, RLHF/RLAIF, RLVR & verifiable-reward design, knowledge distillation, LoRA/QLoRA/PEFT, Ollama, prefix/KV-cache reuse, speculative decoding, hybrid retrieval + reranking, chunking & embedding strategy
- **Core Engineering & DL:** Python, C/C++, PyTorch, HuggingFace Transformers, Docker, Git, Databricks, Hadoop, Spark, MySQL, Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn; Transformers, GNNs, CNNs, RNNs, Deep RL, Causal Inference, Semi-Supervised & Noisy-Label Learning, Spatio-Temporal Reasoning

Industry Experience

MLE Intern **Adobe** **May 2026 – Aug 2026**

Predictive Digital Twin: Agent-Callable Counterfactual Simulation of Marketing Journeys (lead inventor, patent approved by Adobe review committee, in drafting with counsel; research paper in preparation)

[Motivation]: Marketers need to answer what-if questions about how customers would respond to different campaigns, but the most reliable way to find out, 'running A/B tests', is slow, costly, and potentially risky.

- Built a treatment-conditioned generative sequence model (GNN world graph, GRU latent-state customer supernode, treatment cross-attention) that forecasts a customer's full behavioral trajectory under any hypothetical campaign via g-computation rollout, adjusting for past-targeting confounding; downstream metrics like churn, CTR, unsubscribe risk can be derived from the simulated trajectories without task-specific models, so the agent calls one simulation tool instead of orchestrating a separate model per metric.
- Introduced LLM-in-the-loop counterfactual data augmentation that reasons out customer outcomes under unapplied treatments; on a controllable synthetic benchmark with exact counterfactuals, improved individual treatment-effect accuracy by 67-70% over baselines, beating gradient boosting and the Causal/G-Transformer by 36-46%; ablations confirm the augmentation and proposed components drive the gain.
- Validated on real anonymized adobe.com clickstream (matches the performance of the Adobe deployed model in churn prediction) and public OPE/uplift datasets (Open Bandit, Criteo).
- Shipped an **Adobe CoWorker MCP server** that exposes the simulator as agent-callable tools (journey simulation, treatment comparison, audience ranking), letting a marketer drive campaign what-if analysis in natural language; designed typed tool schemas, argument validation, and bounded result payloads so multi-step agent loops stay within context budget, plus a Streamlit demo for non-agent use.

Selected Papers & Research Experience

Research Assistant **North Carolina State University** **Oct 2023 – Present**

Jinming Xing, Muhammad Shahzad, et al. "GRAFT: Generative Reconstruction of Attachment for Text-attributed Cold-start Nodes."

[Motivation]: A cold node arrives with text but zero edges, where every classical link-prediction heuristic degenerates; attaching it to its most text-similar nodes makes aggregation near-identity, collapsing the GNN back into an MLP: similarity-optimal attachment is information-minimal.

- Reframe cold-start classification as posterior inference over latent neighborhoods, marginalizing predictions across sampled neighborhoods instead of committing to one imputed graph, so no single hallucinated edge can

dominate the prediction and attachment uncertainty stays explicit.
- Employ the **LLM as a typed relational verifier** under a bounded per-node call budget: the LLM adjudicates only a 10–20 candidate shortlist into a structured verdict (relation type + confidence) consumed by relation-aware propagation, while bulk retrieval over the full candidate pool stays an embedding job, spending LLM calls only where embeddings are provably inadequate.
- Ship **GRAFT-amortized**, which distills the LLM's relational verdicts on masked training edges into a small cross-encoder, showing the relational signal is genuinely learnable rather than purchased, and giving a second operating point whose accuracy does not ride on an inference-time LLM budget.
- Learn attachment rather than similarity: self-supervised cold-start simulation (edge masking with a degree-stratified curriculum and explicit leakage control), contrastive retrieval with hard negatives drawn from high-similarity non-neighbors, and autoregressive neighborhood construction with a learned STOP action that predicts neighborhood size rather than assuming a fixed top-k.
- Derive an uncertainty signal from disagreement across sampled neighborhoods, enabling principled abstention on ambiguous cold nodes instead of a forced low-confidence label.

Jinming Xing, Muhammad Shahzad, et al. "When Text and Topology Disagree: Bidirectional GNN-LLM Co-Evolution for Noisy Graph Learning." (WSDM'27, Under Review)
[Motivation]: Existing LLM-GNN integration relies on static, unidirectional pipelines, causing unrecoverable error propagation, semantic-structural dissonance, and blind alignment that propagates noise.
- Introduce a Dual-View Co-Evolution mechanism where the GNN injects topological context to guide the LLM, while the LLM constructs a Dynamic Semantic Graph to rewire the GNN: an iterative LLM-in-the-loop feedback cycle that halts mutual error propagation instead of a one-shot pipeline.
- Develop an Uncertainty-Gated Consistency strategy for meta-cognitive alignment: each model acts on the counter-view only when that view is confident: the same confidence-gated decision an agent makes when choosing whether to trust a tool's output or re-plan from scratch.
- Propose a Node-Adaptive Gating Mechanism that learns per-node factors to dynamically fuse static and semantic graphs, resolving the missing link problem by reconstructing connections for isolated nodes.
- Design a Hard-Structure Conflict-Aware Contrastive Loss based on Global Graph Diffusion (PPR), targeting false semantic friends by warping the semantic manifold to respect high-order topological boundaries; add an Entropy-Aware Adaptive Fusion rule for robust inference at test time.
- Outperforming SOTA LLM-GNN integration baselines by up to 9.07% and 7.19% in accuracy and F1-score.

Jinming Xing, Guoheng Sun, et al. "NetSight: Joint Spatio-Temporal Dependency Modeling for Network Traffic Prediction." CIKM'26 (ACM Conf. on Information and Knowledge Management)
[Motivation]: Hidden ST dependencies are ignored. Global-local ST can't be modeled simultaneously.
- A dynamic data-driven spatial-temporal graph was proposed to capture hidden spatial-temporal dependencies; a decoder-only transformer was adopted while GAT serves as the graph convolution backbone.
- Proposed NetSight, which copes with global-local spatial-temporal patterns simultaneously. A simple yet effective learnable pooling was designed, and to speed up the convergence, node normalization was introduced.
- NetSight outperforms the average of baselines by 36.37% (MAE), 32.19% (RMSE), and 23.99% (SMAPE).

Jinming Xing, Muhammad Shahzad. "A Reinforcement Learning Framework for Application-Specific TCP Congestion-Control." IPCCC'25 (IEEE Int'l Performance, Computing and Communications Conf.)
[Motivation]: Diverse optimization goals are overlooked. Client computational load is high.
- Proposed ASC$_{RL}$, a DRL-based framework that can accommodate clients with various optimization goals, requiring per-client reward design rather than a single fixed objective.
- Designed a client-server updating mechanism that decouples training from inference, training stays on the server while only inference runs on the client, reducing the client computational load by up to 91% and making it suitable for deployment on resource-constrained client devices.
- Introduced a fast retraining mechanism for client goals changing, resulting in a 50% reduced retraining time.

Research Assistant **Shenzhen University** **Jun 2021 – Jul 2023**

Jinming Xing, Can Gao, et al. "Weighted fuzzy rough sets-based tri-training and its application to medical diagnosis." Applied Soft Computing [Q1, IF:7.28] (2022)
[Motivation]: Noisy data impedes the learning. Tri-training suffers from the initialization problem.
- Proposed the 'bad-point' technique for dataset de-noising, a high-order information extraction strategy, and three modal data views ('ORI', 'PCA', 'DIS') to initialize tri-training base classifiers.

- The proposed WFRS-Tri framework outperforms baselines under various noisy supervised and semi-supervised scenarios with a range of accuracy improvements from 3% to 15% on average.

Grants

- North Carolina State University College of Engineering Travel Grant, 2025, **$1k**
- National College Students Innovation and Entrepreneurship Training Project, 2021, **$2.8k**
- Guangdong Province Science and Technology Innovation Strategy Special Fund Project, 2022, **$2.1k**

Services

Invited Seminars: "Introduction to GNNs and Their Applications", Jharkhand Rai University, India, May 2025
Teaching Assistant: Computer Network, Data Structures and Algorithms, Senior Design
Program Committee: AAAI'27, WWW'25 (Short Paper Track)
Reviewer: NeurIPS'26, AAAI'26, ICDM'26, WWW'25 (Main Track), TMM, TKDD, TMC, Pattern Recognition, IJCNN'25/26, Soft Computing, Knowledge-Based Systems, Expert Systems With Applications, IJAR, Applied Soft Computing, NeuroComputing, The Journal of Supercomputing, Results in Engineering, Advanced Engineering Informatics, Computer Communications

Full List of Peer-Reviewed Publications

- **Jinming Xing**, Guoheng Sun, Can Gao, et al. "Looking for Bidding Teammates: A Game-Theoretic Model of Stranger Collusion in Peer Review." [ICLR'27, Under Review]
- **Jinming Xing**, Muhammad Shahzad. "When Text and Topology Disagree: Bidirectional GNN-LLM Co-Evolution for Noisy Graph Learning." [WSDM'27, Under Review]
- Chang Xue, Fang Liu, Jiaye Wang, **Jinming Xing**, Chen Yang. "TAS-GNN: A Status-Aware Signed Graph Neural Network for Anomaly Detection in Bitcoin Trust Systems." ICPR'26.
- **Jinming Xing**, Charlotte Brian. "Governance Outruns Analysis: Incidence and Manipulability of Credential-Gated Reciprocal Review." NeurIPS'26 AI-Native Academia.
- Chang Xue, Youwei Lu, Chen Yang, **Jinming Xing**. "RecMind: LLM-Enhanced Graph Neural Networks for Personalized Consumer Recommendations." ICCE'26.
- Guoheng Sun, Ziyao Wang, Xuandong Zhao, Bowei Tian, Zheyu Shen, Yexiao He, **Jinming Xing**, Ang Li. "Invisible Tokens, Visible Bills: The Urgent Need to Audit Hidden Operations in Opaque LLM Services." ICML'26.
- **Jinming Xing**, Guoheng Sun, Hui Sun, et al. "NetSight: Joint Spatio-Temporal Dependency Modeling for Network Traffic Prediction." CIKM'26.
- **Jinming Xing**, Muhammad Shahzad. "A Reinforcement Learning Framework for Application-Specific TCP Congestion-Control." IPCCC'25.
- Qisen Cheng, **Jinming Xing**, Chang Xue, et al. "Unifying Prediction and Explanation in Time-Series Transformers via Shapley-based Pretraining." CSPA'25.
- **Jinming Xing**, Dongwen Luo, Qisen Cheng, et al. "Multi-view Fuzzy Graph Attention Networks for Enhanced Graph Learning." IWPR'25.
- **Jinming Xing**, Chang Xue, Dongwen Luo, et al. "FGATT: A Robust Framework for Wireless Data Imputation Using Fuzzy Graph Attention Networks and Transformer Encoders." ECCS'25.
- **Jinming Xing**, Zhaomin Xiao, Yingyi Wu, et al. "Network Traffic Forecasting via Fuzzy Spatial-Temporal Fusion Graph Neural Networks." ISCMI'24.
- **Jinming Xing**, Qisen Cheng, Dongwen Luo, et al. "Comparative Analysis of Pooling Mechanisms in LLMs: A Sentiment Analysis Perspective." ISMSI'25.
- **Jinming Xing**, Ruilin Xing, Qisen Cheng, et al. "Enhancing Link Prediction with Fuzzy Graph Attention Networks and Dynamic Negative Sampling." ICMSSP'25.
- Can Gao, Jie Zhou, **Jinming Xing**, et al. "Parameterized maximum-entropy-based three-way approximate attribute reduction." International Journal of Approximate Reasoning 151 (2022): 85-100. [Q2, IF: 3.0]
- **Jinming Xing**, Can Gao, et al. "Weighted fuzzy rough sets-based tri-training and its application to medical diagnosis." Applied Soft Computing 128 (2022): 109467. [Q1, IF: 7.28]